Nevertheless, in any real-world application you'll likely need to be able to update your dataset, so we are going to describe how to remove elements: figures 9.13 and 9.14 shows the `remove` method in action on our example k-d tree, and the result of removing point `D` from it.

Similarly to `insert` and `search`, this method is based on binary search tree removal. There are, however, two issues that makes k-d tree's version sensibly more complicated, and they are both connected to how we find a replacement in the tree for the node we are going to remove.

To see what these issues are, we need to take a step back. In binary search trees, when we delete a node, we can face one of tree situations (see figure 9.15):

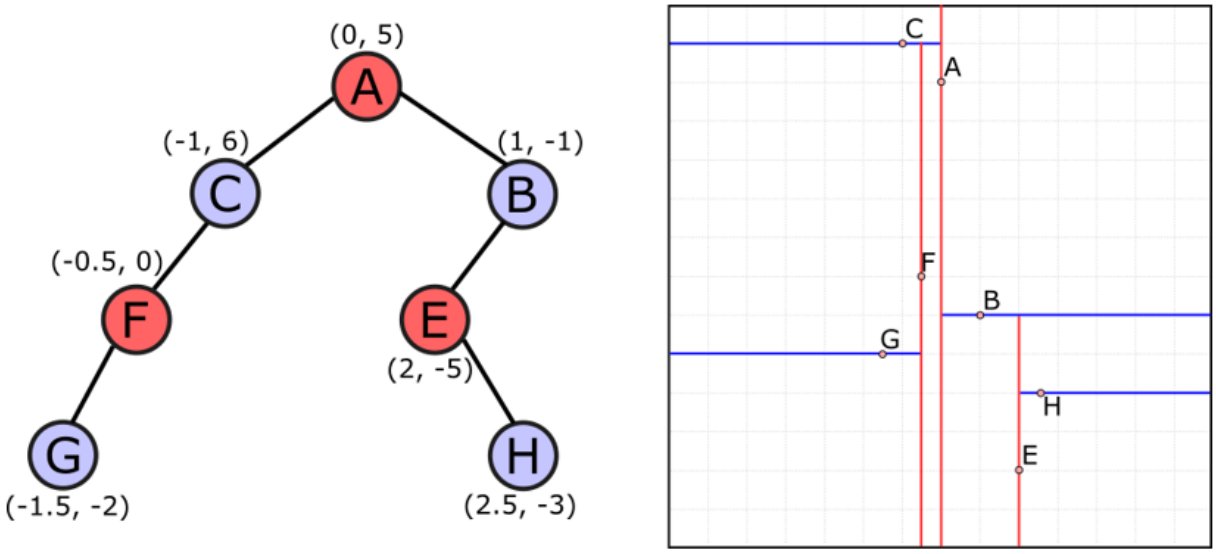


Figure 9.14 The k-d tree resulting after deleting point `D`.

1. The node we need to remove is a leaf: in this situation, we can just safely remove the node from the tree, and